PLANAR STRICTLY CONVEX HYPERRIGIDITY

Abstract. Let K be a compact convex set of affine dimension at most two and let $g\colon K \to \mathbb{R}$ be continuous and strictly convex. We prove that a normalized positive-operator-valued measure and a projection-valued measure on K are equal whenever they agree on every affine function and on g. Equivalently, the function system generated by the affine functions and g is hyperrigid in $C(K)$. Consequently, equality under compression for one such g forces the compression subspace to reduce the entire commuting self-adjoint tuple. No smoothness of g, finite-rank assumption, separability assumption on the acting Hilbert space, or atomicity of the spectral measure is required.

1. Introduction

Let K be a compact convex subset of a finite-dimensional real affine space. Write $A(K)$ for the real affine functions on K, regarded as a self-adjoint function system in $C(K)$. Given a continuous strictly convex function $g\colon K \to \mathbb{R}$, consider

$$S_g = \operatorname{span}_{\mathbb{C}}\big(A(K) \cup \{g\}\big).$$

Scalar strict convexity makes every point of K a Choquet boundary point for S_g. Operator hyperrigidity is stronger: it requires uniqueness not only for scalar representing measures, but for positive-operator-valued measures representing arbitrary, possibly diffuse, spectral representations.

Our main result establishes this operator uniqueness in affine dimension at most two.

Theorem 1.1. *Let K be compact and convex with $\dim \operatorname{aff} K \le 2$. Let E be a normalized positive-operator-valued measure on K, and let F be a projection-valued measure on K, acting on the same Hilbert space. Let $g\colon K \to \mathbb{R}$ be continuous and strictly convex. Suppose*

$$\int_K \ell\, dE = \int_K \ell\, dF \quad (\ell \in A(K)), \qquad \int_K g\, dE = \int_K g\, dF.$$

Then $E = F$.

Here and below a positive-operator-valued measure (POVM) is countably additive in the weak operator topology. Its scalarizations on a compact metric space are finite regular Borel measures. A projection-valued measure (PVM) is a POVM whose values are orthogonal projections.

The corresponding compression statement answers the motivating two-variable question affirmatively.

Corollary 1.2. *Let $X, Y \in B(\mathcal{K})$ be commuting positive contractions, let P be an orthogonal projection, and put*

$$H = \operatorname{ran} P, \qquad A = PXP|_H, \qquad B = PYP|_H.$$

Assume that A and B commute. If $f\colon [0,1]^2 \to \mathbb{R}$ is continuous and strictly convex and

$$Pf(X,Y)P|_H = f(A,B),$$

then $PX = XP$ and $PY = YP$.

The proof has two distinct components. The first is a local interior argument. Supporting-plane defects of g give nonnegative Bregman functions. Douglas factorization would turn any off-diagonal spectral coupling into a nonzero scalar measure of finite reciprocal-Bregman energy. Minty's parametrization of the subgradient graph and a critical inverse-square estimate show that such a measure cannot exist in dimensions one or two.

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The second component treats the boundary. Flat boundary faces are exposed affinely and reduced to the one-dimensional theorem. The remaining extreme boundary may be curved and may carry diffuse spectral measure. We replace each point of K measurably by an extreme-point representing measure and use Scherer's planar extreme-boundary rigidity theorem [7]. Strict Jensen equality then shows that this boundaryization moved no mass. Positivity finally glues the interior, flat faces, and extreme boundary and annihilates all cross terms.

The dimension-one part of Theorem 1.1 is already contained in Brown's one-variable strict-convexity theorem [1]. Known genuinely planar special cases include positive-definite quadratic functions, by Shankar's normal-generator theorem [9], and certain explicitly specified monomial systems in a single normal generator covered by Pietrzycki–Stochel [6]. Davidson–Kennedy [2] give the general relationship among Choquet order, unique extension, and hyperrigidity for function systems.

There is a useful convex-geometric reformulation. Set

$$\Gamma_g = \{(x, g(x)) : x \in K\}, \qquad L_g = \operatorname{conv} \Gamma_g.$$

Restriction to Γ_g identifies $A(L_g)$ completely order isomorphically with S_g, and strict Jensen equality gives $\operatorname{ex} L_g = \Gamma_g$. Consequently,

$$S_g \text{ is hyperrigid in } C(K) \quad \iff \quad A(L_g) \text{ is hyperrigid in } C(\operatorname{ex} L_g).$$

If g is nonaffine, then $\dim \operatorname{aff} L_g = \dim \operatorname{aff} K + 1$. Thus Scherer's planar theorem already handles the graph lift when $\dim \operatorname{aff} K = 1$, whereas an original planar K produces a three-dimensional graph hull. The scope not covered by those results is precisely arbitrary continuous strictly convex g on such two-dimensional K, including diffuse representations. Two separately run solution-aware literature searches completed on July 24, 2026 located no prior theorem with that full scope; this negative search is not a claim of historical priority.

Scherer's later spectrahedra theorem [8] proves hyperrigidity of $A(L)$ in $C(\operatorname{ex} L)$ for compact spectrahedra L with closed extreme boundary. Since $\operatorname{ex} L_g = \Gamma_g$ is already compact, it applies here precisely when the graph hull L_g is a spectrahedron. An arbitrary graph hull of a continuous strictly convex function need not be a spectrahedron. Thus that result is closely related but does not subsume Theorem 1.1.

2. Interior localization

We identify $\operatorname{aff} K$ with $\mathbb{R}^d$, where $d \le 2$, and take all subgradients relative to the single common domain K. For $z \in \operatorname{ri} K$, set

$$\partial_K g(z) = \left\{ p \in \mathbb{R}^d : g(t) \ge g(z) + \langle p, t - z \rangle \text{ for every } t \in K \right\}.$$

Lemma 2.1 (Compact-fiber selection). *Let X, Y be compact metric spaces and let $\mathcal{R} \subset X \times Y$ be closed with every fiber $\mathcal{R}_x = \{y : (x, y) \in \mathcal{R}\}$ nonempty. There is a Borel map $s \colon X \to Y$ such that $s(x) \in \mathcal{R}_x$ for every x.*

Proof. Choose a dense sequence (q_j) in Y and put $d_j(x) = \operatorname{dist}(q_j, \mathcal{R}_x)$. Each d_j is lower semicontinuous. Indeed, if $x_m \to x$, pass to a subsequence realizing the lower limit and choose minimizers $y_m \in \mathcal{R}_{x_m}$. A further subsequence has limit $y \in \mathcal{R}_x$, because $\mathcal{R}$ is closed, and hence

$$d_j(x) \le d(q_j, y) = \liminf_m d_j(x_m).$$

Set $\varepsilon_n = 2^{-n}$. Let $s_1(x)$ be the first q_j with $d_j(x) < \varepsilon_1$. Once s_n is defined, let $s_{n+1}(x)$ be the first q_j satisfying

$$d_j(x) < \varepsilon_{n+1}, \qquad d(q_j, s_n(x)) < \varepsilon_n + \varepsilon_{n+1}.$$

Such a point exists: choose $a \in \mathcal{R}_x$ within ε_n of $s_n(x)$, then choose q_j within ε_{n+1} of a. Every s_n is Borel because it selects the first index satisfying Borel conditions. The displayed estimate makes $(s_n(x))$ Cauchy, and its pointwise limit $s(x)$ is Borel. Since $\operatorname{dist}(s_n(x), \mathcal{R}_x) < \varepsilon_n$ and the fiber is closed, $s(x) \in \mathcal{R}_x$. □

Lemma 2.2. *If $C \subset \operatorname{ri} K$ is compact, then the sets $\partial_K g(z)$, $z \in C$, are nonempty, their union is bounded, their graph is compact, and there is a bounded Borel selector $p(z) \in \partial_K g(z)$.*

Proof. A supporting hyperplane to the epigraph of g at $(z, g(z))$ has nonzero vertical component because $z \in \operatorname{ri} K$; division by that component gives a subgradient. Choose $r > 0$ such that $z + ru \in K$ for $z \in C$ and $\|u\| \leq 1$. If $p \in \partial_K g(z)$, applying the subgradient inequality at $z \pm ru$ gives

$$|\langle p, u\rangle| \leq \frac{\max_K g - \min_K g}{r},$$

so the union of the subdifferentials is bounded.

The graph is closed by passage to the limit in the defining inequalities, and hence compact after restricting the second coordinate to the compact ball supplied by the preceding bound. Lemma 2.1 provides the required Borel selector. □

For such a selector define the supporting defect

$$D_z(t) = g(t) - g(z) - \langle p(z), t - z\rangle .$$

It is nonnegative, and strict convexity implies $D_z(t) = 0$ only when $t = z$. If $C \subset \operatorname{ri} K$ and $J \subset K$ are disjoint compact sets, compactness of the full subgradient graph therefore yields a uniform constant $c > 0$ such that

$$D_z(t) \geq c \qquad (z \in C,\ t \in J). \tag{1}$$

Lemma 2.3 (Critical reciprocal energy)**.** *Let $C \subset \operatorname{ri} K$ be compact and let μ be a nonzero finite Borel measure on C. Then*

$$\iint_{C\times C} \frac{d\mu(t)\, d\mu(z)}{D_z(t)} = +\infty,$$

where $1/0 = +\infty$.

Proof. If $d = 0$, then K, and hence the nonempty support set C, consists of one point z_0. Since $D_{z_0}(z_0) = 0$ and $\mu(C) > 0$, the double integral is $+\infty$. Assume henceforth that $d \in \{1, 2\}$.

Subgradient monotonicity gives

$$D_z(t) + D_t(z) = \langle p(t) - p(z), t - z\rangle \geq 0.$$

Consequently, with $G(z) = (z, p(z))$,

$$D_z(t) \leq \|p(t) - p(z)\|\, \|t - z\| \leq \frac{1}{2}\|G(t) - G(z)\|^2. \tag{2.1}$$

Introduce the Minty variables

$$U(z) = z + p(z), \qquad W(z) = z - p(z).$$

Monotonicity gives $\|W(z) - W(t)\| \leq \|U(z) - U(t)\|$. Hence U is injective, W is a 1-Lipschitz function of U, and

$$\|G(z) - G(t)\|^2 = \frac{1}{2}\big(\|U(z) - U(t)\|^2 + \|W(z) - W(t)\|^2\big) \leq \|U(z) - U(t)\|^2. \tag{2.2}$$

Choose $r_0 \in (0, 1]$ and $M < \infty$ such that, for every $0 < r \leq r_0$, the bounded set $U(C) \subset \mathbb{R}^d$ meets at most Mr^{-d} half-open grid cubes of diameter at most r. Pulling this partition back through U and applying Cauchy–Schwarz gives, with $m = \mu(C) > 0$ and $0 < r \leq r_0$,

$$(\mu \times \mu)\{\|G(z) - G(t)\| \leq r\} \geq \frac{m^2}{M} r^d. \tag{2.3}$$

The layer-cake identity and Tonelli's theorem now yield

$$\iint \frac{d\mu(z)d\mu(t)}{\|G(z) - G(t)\|^2} = 2\int_0^\infty r^{-3}(\mu \times \mu)\{\|G(z) - G(t)\| \leq r\}\, dr \geq \frac{2m^2}{M}\int_0^{r_0} r^{d-3}\, dr.$$

The last integral diverges for $d = 1, 2$. Equation (2.1) completes the proof. □

We use the following standard orientation of Douglas's factorization lemma: if $SS^* \leq TT^*$, then $S = TC$ for some contraction C [3].

Lemma 2.4 (Common-domain localization)**.** *Assume the moment equalities in Theorem 1.1. If $C \subset \operatorname{ri} K$ and $J \subset K$ are compact and disjoint, then*

$$F(C)E(J)F(C) = 0.$$

Proof. Let $R = F(C)$, $T = RE(J)R$, and let $c > 0$ be as in (1). Since D_z is a linear combination of g and affine functions,

$$cE(J) \le \int_K D_z \, dE = \int_K D_z \, dF.$$

On $R\mathcal{H}$, put $M_z = R(\int_K D_z \, dF)R$. Then $cT \le M_z$.

Suppose $T \ne 0$, and choose $u \in R\mathcal{H}$ with $\eta = T^{1/2}u \ne 0$. Douglas factorization gives a contraction C_z such that

$$\sqrt{c}\, T^{1/2} = M_z^{1/2} C_z.$$

Set $v_z = C_z u$ and define the nonzero finite measure $\mu(S) = \|F(S)\eta\|^2$, which is supported on C.

For

$$h_n(s) = \begin{cases} \min\{n, s^{-1}\}, & s > 0, \\ n, & s = 0, \end{cases}$$

functional calculus gives

$$\int_C h_n(D_z(t)) \, d\mu(t) = \frac{1}{c} \langle M_z h_n(M_z) v_z, v_z \rangle \le \frac{1}{c} \|u\|^2.$$

Monotone convergence therefore yields

$$\int_C \frac{d\mu(t)}{D_z(t)} \le \frac{1}{c} \|u\|^2 \qquad (z \in C).$$

The kernel is jointly Borel, so integration in z and Tonelli imply finite reciprocal-Bregman energy, contradicting Lemma 2.3. Thus $T = 0$. □

Lemma 2.5 (Affine exposure)**.** *Let $r \ge 0$ be continuous on K, suppose $\int r \, dE = \int r \, dF$, and set $L = \{r = 0\}$. Then*

$$F(L)E(K \setminus L)F(L) = 0.$$

Proof. For $U_n = \{r \ge 1/n\}$, positivity gives

$$E(U_n) \le n \int r \, dE = n \int r \, dF.$$

Compression by $F(L)$ gives zero. Since $U_n \uparrow K \setminus L$, strong monotone convergence proves the claim. □

Lemma 2.6 (Interval reconstruction)**.** *Let $I = [a, b]$. If a normalized POVM E and a PVM F on I agree on* 1*, the affine coordinate, and a continuous strictly convex function g, then $E = F$.*

Proof. If $a = b$, then I is a singleton. Its only Borel sets are $\varnothing$ and I, and normalization gives $E(I) = F(I) = I$; hence $E = F$. Assume henceforth that $a < b$.

Apply Lemma 2.5 to $t - a$ and $b - t$. Thus the $F(\{a\})$-compression of E is supported at a, and the $F(\{b\})$-compression is supported at b.

Fix a compact $C \subset (a, b)$ and put $R = F(C)$. Exhaust $I \setminus C$ by compact sets disjoint from C and apply Lemma 2.4; this gives $RE(I \setminus C)R = 0$, hence $RE(C)R = R$. In the reverse order, exhaust $(a, b) \setminus C$ by compact sets and obtain

$$\big(F((a, b)) - R\big)E(C)\big(F((a, b)) - R\big) = 0.$$

Endpoint exposure gives the same zero compression on $F(\{a\})$ and $F(\{b\})$. The positivity implications

$$0 \le T \le I, \quad RTR = R \Rightarrow TR = R, \qquad RTR = 0 \Rightarrow TR = 0$$

therefore show that $E(C)$ is the identity on $R\mathcal{H}$ and zero on its orthogonal complement. Thus $E(C) = F(C)$.

Regularity and polarization extend equality to all Borel subsets of (a, b). Hence $E((a, b)) = F((a, b))$. Applying the same positivity implications to the two exposed endpoint projections identifies $E(\{a\})$ and $E(\{b\})$ with their F-counterparts and removes their cross terms. Therefore $E = F$. □

3. The planar boundary

We now assume $\dim \operatorname{aff} K = 2$, identify the affine hull with $\mathbb{R}^2$, and write $K^\circ = \operatorname{ri} K$.

Lemma 3.1 (Boundary stratification)**.** *There is an at most countable family of nondegenerate exposed line segments $L_n \subset \partial K$ such that, with $U_n = \operatorname{ri} L_n$,*

$$K = K^\circ \bigsqcup \operatorname{ex} K \bigsqcup \left(\bigsqcup_{n \geq 1} U_n \right). \tag{2}$$

The endpoints of every L_n are extreme, and $\operatorname{ex} K$ is compact.

Proof. A nonextreme boundary point lies in an open segment $(y, z) \subset K$. Any supporting affine functional at that point vanishes at y and z, so the point belongs to the relative interior of a nondegenerate exposed face. In the plane such a proper face is a line segment. Taking the maximal ones gives precisely the nonextreme boundary.

The relative interior of a maximal boundary segment is relatively open in ∂K: a triangle formed from a smaller subsegment and an interior point fills a one-sided neighborhood of each point of that subsegment. Distinct maximal face interiors are disjoint. Since ∂K is second countable, there are at most countably many. Maximality makes their endpoints extreme, and

$$\operatorname{ex} K = \partial K \setminus \bigcup_n U_n$$

is closed in the compact boundary. □

Put $Z = \operatorname{ex} K$ and let $\mathcal{P}(Z)$ carry the weak topology.

Lemma 3.2 (Measurable extreme representation)**.** *There is a Borel map $x \mapsto \nu_x \in \mathcal{P}(Z)$ satisfying*

$$\int_Z z \, d\nu_x(z) = x, \qquad \nu_z = \delta_z \quad (z \in Z). \tag{3.2}$$

Proof. Every boundary point is either extreme or lies on some L_n, and hence is a convex combination of at most two points of Z. Intersecting a line through an interior point with K shows that every point of K is a convex combination of at most four points of Z.

The barycenter map $b \colon \mathcal{P}(Z) \to K$ is continuous. Thus

$$\mathcal{R} = \{(x, \nu) : b(\nu) = x\}$$

is a closed subset of the compact metric space $K \times \mathcal{P}(Z)$, with nonempty compact fibers. Lemma 2.1 gives a Borel selector $x \mapsto \nu_x$.

To prove the last assertion, let a probability measure ν have barycenter $z \in Z$, and suppose $\nu \neq \delta_z$. Some real linear functional a is then nonconstant ν-almost surely. After changing its sign if necessary, the set $A = \{t : a(t) > a(z)\}$ has positive measure; the barycenter identity makes its complement have positive measure as well. If y_A, y_B are the barycenters of the two normalized restrictions and $\alpha = \nu(A)$, then

$$z = \alpha y_A + (1 - \alpha) y_B, \qquad 0 < \alpha < 1,$$

while $a(y_A) > a(z)$. This is a nontrivial convex decomposition of z, contrary to extremality. Hence $\nu_z = \delta_z$. □

For a normalized POVM G on K, define its boundaryization on Z by

$$\widehat{G}(S) = \int_K \nu_x(S) \, dG(x) \qquad (S \in \mathfrak{B}(Z)). \tag{3.3}$$

The evaluation map $\nu \mapsto \nu(S)$ is Borel for Borel S; see, for example, the standard probability-kernel machinery in [4]. For $\xi, \eta \in \mathcal{H}$, scalarize G by $G_{\xi,\eta}(S) = \langle G(S)\xi, \eta \rangle$. The usual bounded-kernel Fubini identity applies first to the positive diagonal measures $G_{\xi,\xi}$; polarization then gives it for $G_{\xi,\eta}$. Scalar monotone convergence proves countable additivity of $\widehat{G}$, and normalization follows from $\nu_x(Z) = 1$. Thus $\widehat{G}$ is a POVM, and for every bounded Borel h the weak-operator integrals satisfy

$$\int_Z h\, d\widehat{G} = \int_K \left(\int_Z h\, d\nu_x \right) dG(x). \tag{3}$$

We use the following form of Scherer's planar theorem [7, Theorem 3.8]. In Scherer's formulation, $A(K)$ is hyperrigid in $C(Z)$. Indeed, a PVM N on Z induces a representation $\rho(h) = \int_Z h\, dN$, while a normalized POVM M induces a unital completely positive map $\Phi(h) = \int_Z h\, dM$. Agreement of their affine moments says $\Phi|_{A(K)} = \rho|_{A(K)}$. Hyperrigidity gives $\Phi = \rho$, and scalar regular-measure uniqueness followed by polarization gives $M = N$. We use here the standard POVM–UCP correspondence for commutative C^*-algebras; see [5]. Hence the following POVM/PVM form is exactly the consequence required here.

Theorem 3.3 (Scherer)**.** *Let M be a normalized POVM and N a PVM on $Z = \operatorname{ex} K$. If*

$$\int_Z \ell\, dM = \int_Z \ell\, dN \qquad (\ell \in A(K)),$$

then $M = N$.

Lemma 3.4 (Extreme-supported reconstruction)**.** *Let G be a normalized POVM on K and let N be a PVM supported on Z. If G and N agree on every affine function and on g, then $G = N$.*

Proof. By the barycenter identity and (3), $\widehat{G}$ and N agree on every affine function. Hence Theorem 3.3 gives $\widehat{G} = N$.

Define

$$\widetilde{g}(x) = \int_Z g(z)\, d\nu_x(z).$$

Jensen's inequality gives $\widetilde{g} \geq g$. Equality holds precisely on Z. Indeed, if a probability measure ν is not a point mass and S, T are independent with law ν, strict convexity is strict on the positive-probability event $S \neq T$, and

$$g\left(\int z\, d\nu(z) \right) \leq \mathbb{E} g\left(\frac{S+T}{2} \right) < \mathbb{E} g(S) = \int g\, d\nu.$$

For $z \in Z$, the identity $\nu_z = \delta_z$ gives $\widetilde{g}(z) = g(z)$.

The g-moment equality and $\widehat{G} = N$ now imply

$$\int_K (\widetilde{g} - g)\, dG = 0.$$

For $S_m = \{\widetilde{g} - g \geq 1/m\}$, positivity gives $G(S_m) = 0$. Since $K \setminus Z = \bigcup_m S_m$, the POVM G is supported on Z. Boundaryization is therefore the identity on G, so $G = \widehat{G} = N$. ☐

4. Proof of the main theorem

Proof of Theorem 1.1. Affine dimension zero is immediate, and affine dimension one is Lemma 2.6. Assume therefore that $\dim \operatorname{aff} K = 2$, and use the boundary stratification (2). Write its strata as

$$D_0 = K^\circ, \qquad D_* = \operatorname{ex} K, \qquad D_n = U_n.$$

Let

$$R_0 = F(D_0), \qquad R_* = F(D_*), \qquad R_n = F(D_n).$$

These projections are pairwise orthogonal and have strong sum I.

For each stratum define

$$E_i(S) = R_i E(S) R_i, \qquad F_i(S) = F(S \cap D_i).$$

Compression preserves all affine and g-moment equalities.

We first reconstruct the interior. If $C \subset K^\circ$ is compact, put $Q = F(C)$ and $Q' = R_0 - Q$. Exhaust $K \setminus C$ and $K^\circ \setminus C$ by compact sets and apply Lemma 2.4 in the two orders. Strong monotone convergence gives

$$QE_0(C)Q = Q, \qquad Q'E_0(C)Q' = 0.$$

Since $0 \le E_0(C) \le R_0$, positivity forces $E_0(C) = Q$. Regularity and polarization extend this to every Borel subset of K°; thus $E_0 = F_0$.

For a flat face L_n, choose an affine $r_n \ge 0$ with $L_n = \{r_n = 0\}$. Lemma 2.5 first forces the $F(L_n)$-compression of E to be supported on L_n. On that interval the compressed measures agree on 1, an affine coordinate, and $g|_{L_n}$, which remains strictly convex. Lemma 2.6 therefore identifies them. Compression to $R_n = F(U_n)$ gives $E_n = F_n$.

On $R_*\mathcal{H}$, the PVM F_* is supported on $Z = \operatorname{ex} K$. Lemma 3.4 gives $E_* = F_*$.

It remains to remove cross terms. Fix a Borel set S and put

$$T = E(S), \quad Q_i = F(S \cap D_i), \quad P_i = F(D_i \setminus S).$$

The diagonal identities imply

$$Q_iTQ_i = Q_i, \qquad P_iTP_i = 0.$$

Because $0 \le T \le I$, positivity yields $TQ_i = Q_i$ and $TP_i = 0$. Taking strong joins over all strata gives

$$TF(S) = F(S), \qquad TF(K \setminus S) = 0.$$

Since the two spectral projections sum to I, $T = F(S)$. Thus $E = F$. □

Remark 4.1. The proof uses the inverse-square energy only on parameter spaces of affine dimension at most two. It does not assert the corresponding theorem in unrestricted affine dimension three.

5. Compression and hyperrigidity consequences

Corollary 5.1. *Let $K \subset \mathbb{R}^m$ be compact and convex with $\dim \operatorname{aff} K \le 2$. Let $T_1, \dots, T_m$ be commuting bounded self-adjoint operators with joint spectrum in K. Let P be an orthogonal projection, put $H = \operatorname{ran} P$ and*

$$A_j = PT_jP|_H,$$

and assume the A_j commute. If $g \colon K \to \mathbb{R}$ is continuous and strictly convex and

$$Pg(T_1, \dots, T_m)P|_H = g(A_1, \dots, A_m),$$

then P reduces every T_j.

Proof. Let $V \colon H \to \mathcal{K}$ be the inclusion. First, $\sigma(A_1, \dots, A_m) \subset K$. Indeed, if an affine function ℓ is nonnegative on K, then

$$\ell(A) = V^*\ell(T)V \ge 0.$$

A joint spectral point outside K would be strictly separated from K by such an affine ℓ, contradicting positivity of $\ell(A)$.

Compress the joint spectral PVM of $T = (T_1, \dots, T_m)$ to obtain a POVM E, and let F be the joint spectral PVM of $A = (A_1, \dots, A_m)$. They agree on affine functions and on g, so Theorem 1.1 gives $E = F$. In particular,

$$V^*T_j^2V = A_j^2.$$

But

$$V^*T_j^2V - A_j^2 = V^*T_j(I-P)T_jV = \big((I-P)T_jV\big)^*\big((I-P)T_jV\big).$$

Thus $(I-P)T_jP = 0$; taking adjoints gives the other off-diagonal block, so $PT_j = T_jP$. □

Corollary 1.2 is the case $m = 2$ and $K = [0,1]^2$.

Corollary 5.2. *For compact convex K of affine dimension at most two and continuous strictly convex g, the function system S_g is hyperrigid in $C(K)$.*

Proof. A unital completely positive extension of a representation of $C(K)$ corresponds to a normalized POVM, while the representation corresponds to its PVM. Agreement on S_g is exactly agreement on the affine functions and on g. Theorem 1.1 makes the two measures equal. The affine functions contain the constants and separate the points of K, so Stone–Weierstrass gives $C^*(S_g) = C(K)$. Moreover, K is compact metrizable because it lies in a finite-dimensional affine space; hence $C(K)$ is separable. The standard unique-extension characterization of hyperrigidity therefore applies, with no separability assumption on the acting Hilbert space; see [2, 6]. □

Verification and provenance status

The ordinary proof was generated with OpenAI GPT-5.6 Pro. OpenAI Codex supported manuscript preparation, audit coordination, provenance tooling, and release infrastructure. DannyExperiments initiated, curated, validated, and published the project.

The ordinary proof underlying this manuscript was frozen before review and subjected to multiple separately run AI-assisted hostile audits. An initial assembled-manuscript audit passed Theorem 1.1 and its corollaries while identifying the zero-dimensional and singleton omissions repaired in Lemmas 2.3 and 2.6. A subsequent audit of the repaired manuscript passed all fourteen numbered theorem, lemma, and corollary statements but classified the submission overall as repairable because of documentary, citation, provenance, and release-wording defects addressed in this version. These audits are not human peer review, and specialist review remains pending.

The returned Aristotle feasibility project reports a successful build under Lean/mathlib v4.28.0. Repository scans find no `sorry`, `admit`, new mathematical axiom, or unsafe escape in its foundational POVM/PVM files. Its declaration `thm_main` matches the manuscript theorem, while `cor_tuple` and `cor_square` are stronger ambient-space surrogates: they place the compressed tuple and its unital functional calculus on the full Hilbert space rather than on $\operatorname{ran} P$. All three declarations remain explicit `sorry` declarations, and independent replay remains pending. Accordingly, this manuscript does not claim that its main theorem or operator corollaries are formally verified. The source artifacts, audit reports, literature reports, Aristotle return, checksums, and build workflows are preserved in a research repository at `https://github.com/DannyExperiments/planar-strict-convex-hyperrigidity`. Two separately run solution-aware literature searches found the dimension-one case in prior work and located no prior theorem with the full scope of Theorem 1.1; this remains evidence from a negative search, not a claim of historical priority.

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